Year 1 AS Pure Maths

Chapter 14

Exponentials and logarithms

Practice Booklet

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14.1 Exponential functions

Practice Book, printed page 113–114.

1. Write each of the following as exact fractions:

a. $(\frac{2}{3})^3$

b. 4^{-2}

c. $(\frac{1}{5})^{-2}$

d. $(\frac{4}{3})^{-3}$

Hint

$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
$a^{-m} = \frac{1}{a^m}$ ← Sections 1.1, 1.4

2.

a. Copy and complete the table for $y = 1.2^x$, giving each answer to 2 decimal places.

x	-3	-2	-1	0	1	2	3
y							

b. Use the table to draw an accurate graph of $y = 1.2^x$.

Hint

Use your calculator to work out each y-value in the table. Then plot the points.

3.

a. Copy and complete the table for $y = (\frac{2}{3})^x$, giving each answer as an exact fraction.

x	-3	-2	-1	0	1	2	3
y							

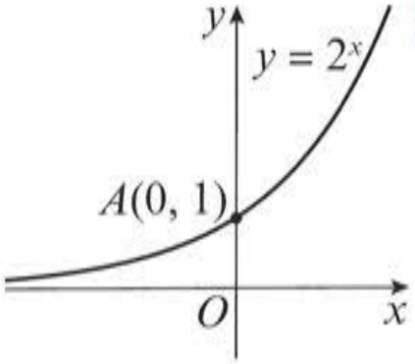
b. Use the table to draw the graph of $y = (\frac{2}{3})^x$.

c. Use your graph to solve $(\frac{2}{3})^x = 2$.

Hint

Draw the line $y = 2$ onto your graph. Read off the x-coordinate of the point of intersection.

4. $f(x) = 2^x$
The diagram shows a sketch of the graph $y = f(x)$.



Use the graph to draw a sketch of the following, labelling the new coordinates of point A and any points of intersection with the coordinate axes:

a. $y = f(x + 2)$
b. $y = f(x) - 2$
c. $y = 2f(x)$
d. $y = -f(x)$

Hint

If the graph has an asymptote other than one of the coordinate axes, you should draw and label it in your sketch.

5. E $f(x) = (\frac{1}{2})^x$

a. Sketch the graph of $y = f(x)$, labelling any points of intersection with the coordinate axes and giving the equations of any asymptotes. (2 marks)

b. Use your answer to part **a** to sketch (labelling the equation of any asymptotes):

i. $y = f(x + 3) - 4$
ii. $y = -2f(x)$

(4 marks)

6. E $g(x) = (\frac{3}{2})^x$
On the same set of axes, sketch the following two graphs, labelling any points of intersection with the coordinate axes and giving the equation of any asymptotes.

a. $y = g(x)$ and $y = g(-x)$ (2 marks)

b. $y = g(x)$ and $y = g(\frac{1}{2}x)$ (2 marks)

7. E/P The graph of $y = ka^x$ passes through the points $(1, 2)$ and $(-2, 128)$.
Find the values of the constants k and a. (3 marks)

14.2 $y = e^x$

Practice Book, printed page 114.

1. On the same set of axes, sketch the graphs of $y = e^x$ and:

a. $y = e^{-x}$

b. $y = e^{x+5} - 6$

c. $y = -3e^x - 5$

d. $y = e^{4x}$

e. $y = 6 - 5e^x$

f. $y = e^{x/2} - 2$

Hint

You will need six different sketches. Each sketch should show $y = e^x$ and one other function.

2. Find $f'(x)$ where $f(x)$ equals:

a. $e^{x/2}$

b. e^{-5x}

c. $-4e^{6x}$

d. $7e^{x/3}$

Hint

For all real values of x, and for any constant k, if $f(x) = e^{kx}$ then $f'(x) = ke^{kx}$

3. Find $\frac{dy}{dx}$ where y equals:

a. $3e^{2x} + e^{4x}$

b. $6e^{-2x} + \frac{5}{e^x}$

c. $e^{2x}(e^{-x} - 4)$

d. $\frac{1}{e^{3x}}(2 - e^{5x})$

Hint

First expand the brackets or simplify the fraction.
For all real values of x, and for any constant k, if $y = e^{kx}$ then $\frac{dy}{dx} = ke^{kx}$

4. Write each of the following in the form $f(x) = Ae^{bx}$, where A and b are constants whose values are to be found.

a. e^{5x+4}

b. e^{7x-1}

c. $e^{3-x/2}$

Hint

Use the laws of indices: $e^{m+n} = e^m \times e^n$
You can give exact values as powers of e.

5. E Differentiate with respect to x:

a. $8x^2 - \frac{1}{e^{6x}}$ (2 marks)

b. $2e^{-x}(4e^{5x} - 3)$ (3 marks)

6. E Given that $y = 5\sqrt{x} + \frac{10}{x^2} - 4e^{-2x}$, find $\frac{dy}{dx}$ (3 marks)

7. E

a. Show that $e^{2(3x-1)}$ can be written in the form form Ae^{bx}, where A and b are constants whose values are to be found. (2 marks)

b. Hence, or otherwise, sketch the graph of $y = e^{2(3x-1)}$. In your sketch, label any asymptotes and the coordinates of any intercepts of the y-axis. (3 marks)

8. E/P Given $f(x) = e^{4x}$, find the exact equation of the tangent at the point where $x = \frac{1}{2}$ on the curve with equation $y = f(x)$. (4 marks)

14.3 Exponential modelling

Practice Book, printed page 115–116.

1. $P = 20000e^{0.01t}$

a. Find $\frac{dP}{dt}$

b. Find the value of $\frac{dP}{dt}$ when $t = 10$.

Hint

First differentiate and then substitute $t = 10$ into your answer.

2. $V = 32000e^{-0.08t}$

a. Sketch the graph of V against t.

b. Find $\frac{dV}{dt}$

c. Find the value of $\frac{dV}{dt}$ when $t = 20$.

d. Given that V is the value of a car and t is the time since purchase, interpret the significance of the sign of your answer to part **c**.

Hint

A positive slope means that the function is increasing.
A negative slope means that the function is decreasing.

3. The number of deer, D, in a population after t years is modelled by the formula $D = 1300e^{t/8}$

a. Use this model to estimate the number of deer after:

i. 0 years

ii. 2 years

iii. 30 months

b. Use your answer to part **a i** to interpret the meaning of the constant 1300 in the model.

c. Find $\frac{dD}{dt}$

d. Show that after 8 years, the deer population is growing by approximately 440 deer per year.

e. By substituting $t = 80$ into $\frac{dD}{dt}$, explain why this model is not valid for large values of t.

4. E The value, V, of a car, in £s, is given by $V = 29000e^{-0.12t}$, where t is the time in years since the car was purchased.

a. Find the value of the car after 6 years. (1 mark)

b. Find the value of $\frac{dV}{dt}$ when:

i. $t = 2$

ii. $t = 6$

(3 marks)

c. Use you answer to parts **a** and **b** to describe how the value of the car decreases over the first 6 years of ownership. (2 marks)

5. E/P In 1980, the population, P, of a town was 18000. The population in subsequent years can be modelled $P = P_0e^{0.02t}$, where t is the time in years since 1980.

a. State the value of P_0. (1 mark)

b. Find the rate, $\frac{dP}{dt}$, in people per year to the nearest person, at which the population P is increasing in 2005. (3 marks)

c. Explain why this model is not valid for large values of t. (1 mark)

6. E/P A child takes a dose of 250 mg of paracetamol. The level of paracetamol, P, present in the body at time t hours after it is taken can be given by the formula $P = 250e^{-0.15t}$.

a. Use the model to estimate the level present after 6 hours. (1 mark)

b. Show that $\frac{dP}{dt} = kP$, where k is a constant to be found. (2 marks)

c. Interpret the sign of k. (1 mark)

7. E/P The value, V, in £s, of a house t years after it reached a low value due to a property crash, can be modelled by the equation $V = 150000e^{0.06t}$.

a. State the value of the house at time $t = 0$. (1 mark)

b. Use the model to estimate the value of the house after 7 years. (1 mark)

c. Find $\frac{dV}{dt}$ (2 marks)

A new property crash is likely to happen when the value of the house is increasing by £15000 per year.

d. Show that between $t = 8$ and $t = 9$ the conditions are correct for a new crash. (3 marks)

14.4 Logarithms

Practice Book, printed page 116–117.

1. Complete the table by writing the equivalent logarithm or power.

	Logarithm	**Power**
a		$10^3 = 1000$
b	$\log_2 8 = 3$	
c		$7^2 = 49$
d		$6^{-3} = \frac{1}{216}$
e	$\log_4 \frac{1}{256} = -4$	
f	$\log_9 9 = 1$	
g		$25^{1/2} = 5$
h	$\log_{27} 3 = \frac{1}{3}$	

Hint

$a^x = n$ is equivalent to $\log_a n = x$ $(a \neq 1)$

2. Without using a calculator, find the value of:

a. $\log_8 64$

b. $\log_2 \frac{1}{16}$

c. $\log_5 5$

d. $\log_3 \sqrt[3]{3}$

e. $\log_{10} 0.01$

f. $\log_9 1$

Hint

For example, if $\log_8 64 = x$ then $8^x = 64$.

3. Without using a calculator, find the value of x for which:

a. $\log_3 x = 4$

b. $\log_5(3x - 1) = 3$

c. $\log_x 729 = 3$

d. $\log_x(4x) = 2$

Hint

For example, if $\log_3 x = 4$ then $3^4 = x$.

4. Evaluate each of these logarithms to four decimal places.

a. $\log_5 18$

b. $\log_e 0.6$

c. $2\log_8 9$

Hint

Use your calculator.

5. E Find the exact value of x for which $\log_8(2x - 1) = \frac{5}{3}$ (3 marks)

6. E Find the exact value of x for which $\log_x 64 = \frac{3}{4}$ (3 marks)

7. E Solve the equation $\log_{125} x = -\frac{4}{3}$ (3 marks)

8. E/P Find the exact value of $\log_a(\sqrt[3]{a^2})$. (3 marks)

14.5 Laws of logarithms

Practice Book, printed page 117–118.

1. Write as a single logarithm:

a. $4\log_7 2$

b. $\log_3 16 - \log_3 2$

c. $\log_5 6 + \log_5 7$

d. $3\log_4 5 + 2\log_4 8$

e. $3\log_7 4 - 4\log_7 3$

f. $\log_6 3 + \log_6 4 - \log_6 5$

Hint

$\log_a(x^k) = k\log_a x$
$\log_a x + \log_a y = \log_a xy$
$\log_a x - \log_a y = \log_a(\frac{x}{y})$

2. By writing as a single logarithm, evaluate the following without using a calculator:

a. $\log_{10} 5 + \log_{10} 20$

b. $4\log_8 2 + \log_8 4$

c. $\log_2 48 - \log_2 6$

d. $\log_5 4 - \log_5 20$

e. $2\log_8 5 + 4\log_8 2 - 2\log_8 10$

3. Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$:

a. $\log_a a^3$

b. $\log_a(x^2y^3)$

c. $\log_a(\frac{x^3}{z^4})$

d. $\log_a(\frac{x\sqrt{y}}{z^2})$

e. $\log_a(\sqrt[3]{ax^2yz^4})$

f. $\log_a(\frac{a^3x^4y}{\sqrt{z}})$

Hint

Use the power law, multiplication law and division law of logarithms.
Remember that $\log_a a = 1$.

4. Solve the following equations.

a. $\log_6 4 + \log_6 x = 2$

b. $2\log_8 6 = \log_8 x - 1$

c. $2\log_6 x = 2 + \log_6 4$

d. $2\log_4(x-1) - \log_4 9 = 1$

Hint

First move the logarithms onto the same side of the equation and then simplify the equation using either the multiplication or division laws of logarithms.

5. E Solve the equation $3\log_5 4 = 2 - 2\log_5 x$. (3 marks)

6. E/P Solve the equation $\log_5 x + 2\log_5 x = 4$, giving your answer in the form $x = p(\sqrt[q]{r})$, where p, q and r are integers to be found. (3 marks)

7. E Solve the equation $\log_3(2x+5) - \log_3(4x-1) = 2$. (4 marks)

8. E/P

a. Given that $\log_2(x+5) - 2\log_2(x-1) = 1$, show that $2x^2 - 5x - 3 = 0$. (4 marks)

b. Hence, or otherwise, solve $\log_2(x+5) - 2\log_2(x-1) = 1$. (2 marks)

14.6 Solving equations using logarithms

Practice Book, printed page 118–119.

1. Solve, giving your answer to 3 significant figures:

a. $6^x = 31$

b. $3^x = 70$

c. $5^{2x} = 640$

Hint

Rewrite each exponential equation as a logarithmic equation and solve using your calculator.

2. Solve, giving your answer to 3 significant figures:

a. $5^{x+3} = 0.3$

b. $4^{2x-5} = 85$

c. $7^{1-4x} = 11$

Hint

You can use the $\log_\square \square$ button on your calculator to find logs to any base.

3. Solve, giving your answer to 3 significant figures:

a. $5^{2x} - 11(5^x) + 30 = 0$

b. $4^{2x} - 19(4^x) + 60 = 0$

c. $5^{2x+1} - 19(5^x) + 18 = 0$

Hint

Write $5^{2x} = (5^x)^2$. Use the substitution $u = 5^x$ to write a quadratic in terms of u. Then factorise and solve the resulting quadratic. Finally, use logarithms to find x for the given values of 5^x.

4.

a. Solve the quadratic equation $9u^2 - 3u - 20 = 0$.

b. Substitute $u = 3^x$ to show that the equation $3^{2x+2} - 3(3^x) - 20 = 0$ can be written as $9u^2 - 3u - 20 = 0$.

c. Hence solve the equation $3^{2x+2} - 3(3^x) - 20 = 0$, explaining clearly why there is only one real solution.

Hint

Remember that you cannot evaluate the logarithm of a negative number.

5. Solve, giving your answer to 3 significant figures:

a. $8^x = 5^{x+3}$

b. $7^{3-x} = 3^{2x}$

c. $6^{2x} = 4^{x+3}$

Hint

Take the logarithm of both sides and then use the power law of logarithms to solve the resulting equation.

6. E Solve each of the following equations, giving your answers to 3 significant figures where appropriate.

a. $7^{2x-5} = 400$ (2 marks)

b. $6^{2x} + 4(6^x) - 21 = 0$ (3 marks)

7. E Solve each of the following equations, giving your answers to 3 significant figures where appropriate.

a. $4^{x+2} = 5^{x-1}$ (2 marks)

b. $\log_7(5 - x) = 0.8$ (2 marks)

8. E

a. Sketch the graph of $y = -2^x + 7$, stating the coordinates of any point where the graph crosses the coordinate axes and the equations of any asymptotes. Write coordinates as a logarithm when necessary. (3 marks)

b. Solve the equation $-2^x + 7 = -6$. (2 marks)

9. E/P

a. Using the substitution $u = 3^x$, show that the equation $9^x - 3^{x+1} - 10 = 0$ can be written in the form $u^2 - 3u - 10 = 0$. (2 marks)

b. Hence solve the equation $9^x - 3^{x+1} - 10 = 0$, giving your answer to 2 decimal places. (3 marks)

14.7 Working with natural logarithms

Practice Book, printed page 119–120.

1. Find the values of x, giving your answers in the form $a+b\ln c$, where a, b and c are rational constants.

a. $e^x = 8$
b. $e^{x/2} = 6$
c. $e^{x+5} = 81$
d. $e^{3x-1} = 23$
e. $4e^{1-x} = 29$
f. $\frac{2}{3}e^{3x} = 7$

Hint

Take the natural logarithm of both sides: $\ln(e^x) = x$.

2. Solve these equations, giving your answers in exact form.

a. $\ln x = 11$
b. $\ln 6x = 3$
c. $\ln(3-5x) = -1$
d. $\ln(5-2x) = \frac{2}{5}$
e. $2\ln(3x-5) = 7$
f. $\ln(x^2-9x+21) = 0$

Hint

Use the fact that $e^{\ln x} = x$.

3. Solve the equation $e^{2x} - 5e^x + 4 = 0$, giving your answer in the form $\ln k$, where k is an integer to be found.

Hint

Use the substitution $u = e^x$ to form a quadratic in u.
Solve the quadratic in u and hence find the value of x.

4. Solve the equation $2e^{4x} - 11e^{2x} + 14 = 0$, giving your answers in the form $a\ln b$, where a and b are rational constants to be found.

Hint

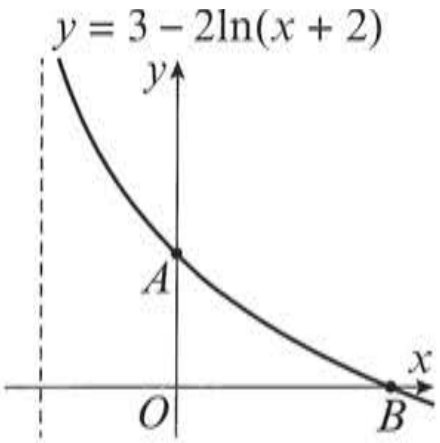
Use the substitution $u = e^{2x}$.

5. E Find the exact solution to the equation $e^x - 12 = e^{x/2}$. (4 marks)

6. E Find the exact solutions of the equation $2e^{4x} + 15e^{-4x} = 13$. (5 marks)

7. E/P Solve $e^{1-6x} = 4(12^x)$, giving your answer in the form $\frac{a+\ln b}{c+\ln d}$ (4 marks)

8. E The graph of $y = 3 - 2\ln(x + 2)$ is shown in the diagram.



Find the exact coordinates, leaving your answer in terms of e or a natural logarithm where necessary, of:

a. A (2 marks)

b. B (3 marks)

9. E/P The temperature, T, in degrees Celsius, of bath water can be modelled by the equation $T = 31e^{-0.05x} + 7$, $x \geq 0$, where x is the time in minutes since the bath tub was full.

a. Sketch a graph of $T = 31e^{-0.05x} + 7$, clearly labelling any points of intersection with the coordinate axes and the equation of any asymptotes. (3 marks)

b. Interpret the meaning of the 7 in the equation. (1 mark)

Jack will add more hot water once the temperature reaches 25°C.

c. Given that Jack enters the bath at $x = 0$, find the time when Jack will first add more hot water. (3 marks)

14.8 Logarithms and non-linear data

Practice Book, printed page 120–122.

1. Two variables V and x satisfy the formula $V = 5x^3$.

a. Show that $\log_{10} V = \log_{10} 5 + 3\log_{10} x$.

b. Sketch the straight line graph of $\log_{10} V$ against $\log_{10} x$, labelling the intercept on the vertical axis.

c. State the gradient of the line.

Hint

Begin by taking logs of both sides of the equation.

2. Two variables H and t satisfy the formula $H = 3(\frac{3}{2})^t$.

a. Show that $\log_{10} H = \log_{10} 3 + t\log_{10} \frac{3}{2}$

b. Sketch the straight line graph of $\log_{10} H$ against t, labelling the intercept on the vertical axis.

c. State the exact gradient of the line.

Hint

Begin by taking logs of both sides of the equation.

3. The data in the table follows a trend of the form $y = ab^x$, where a and b are constants.

x	0.5	2	5	7
y	847.5	1878.5	9229.1	26672.0

a. Show that $\log_{10} y = \log_{10} a + x \log_{10} b$.

b. Copy and complete the table of values of x and $\log y$, giving your answers to 2 decimal places.

x	0.5	2	5	7
$\log y$				

c. Plot the graph of $\log y$ against x and draw in a line of best fit.

d. Use your graph to estimate the values of a and b to 2 significant figures.

4. The value, V, of a car can be modelled using an exponential function $V = ab^t$, where t is the time in years since the car was purchased and a and b are constants.
The car was purchased at the beginning of 2008.
The table shows value at the beginning of subsequent years.

Year	2008	2010	2012	2014	2016	2018
Value, £V	25000	14823	8788	5211	3089	1832

a. Copy and complete the table below. Give your answers to 2 d.p.

Time in years since 2008, t	0	2	4	6	8	10
$\log_{10} V$						

b. Show that $V = ab^t$ can be rearranged into the form $\log_{10} V = \log_{10} a + t \log_{10} b$.

c. Plot the graph of $\log_{10} V$ against t using the values in the table.

d. Draw a line of best fit and calculate the gradient.

e. Use the graph to estimate the values of a and b to 3 significant figures.

5. E/P The price of a stock, P, in pence, of a company can be modelled using an exponential function $P = ab^t$, where t is the time in days since the company was taken over and a and b are constants.
The takeover was complete on Friday 9/3/18 and the new stock began trading on Monday 12/3/18.
The table shows the price of the stock over five days.

Date	12/3/18	13/3/18	14/3/18	15/3/18	16/3/18
Price, P **pence**	27.43	29.76	32.29	35.04	38.01

a. Copy and complete the table below. Give your answers to 2 d.p. (1 mark)

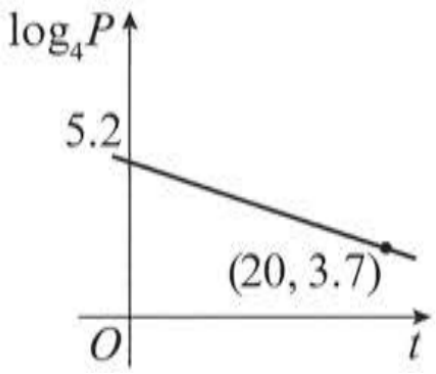
Time in days since 12/3/18, t	0	1	2	3	4
$\log_{10} P$					

b. Show that $P = ab^t$ can be rearranged into the form $\log_{10} P = \log_{10} a + t \log_{10} b$. (2 marks)

c. Plot the graph of $\log P$ against t and draw a line of best fit. (2 marks)

d. Use the graph to estimate the values of a and b to 3 significant figures. (2 marks)

6. E/P The number of people, P, left in a concert venue t minutes after the band stops playing can be modelled by an equation of the form $P = ab^t$. The diagram shows the graph of $\log_4 P$ against t.



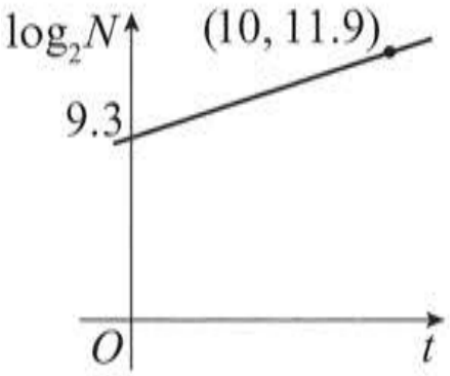
a. Write down the equation of the line shown on the graph. (2 marks)

b. Find the number of people in the venue when $t = 0$, correct to 3 significant figures. (1 mark)

c. Find the values of a and b in the model, correct to 3 s.f. (2 marks)

d. Find the number of people half an hour after the band stopped playing. (1 mark)

7. E/P A scientist recorded the number of bacteria, N, growing in a petri dish after t minutes. The diagram shows a graph of $\log_2 N$ against t.



a. Write down the equation of the line. (2 marks)

A scientist says the data can be modelled by an equation in the form $N = ab^t$

b. Find the values of a and b to 1 decimal place. (4 marks)

c. Interpret the meaning of the constant a in this model. (1 mark)

Problem solving: Set A

Practice Book, printed page 122.

Bronze

Solve the equation $6^{3x-2} = 23$, giving your answer to 2 decimal places. (3 marks)

Silver

a. Using the substitution $u = 3^x$, show that the equation $9^x - 3^{x+2} + 20 = 0$ can be written in the form $u^2 - 9u + 20 = 0$. (2 marks)

b. Hence solve the equation $9^x - 3^{x+2} + 20 = 0$, giving your answer in the form $\log_a b$. (3 marks)

Gold

Find the exact values of x for which $4^{2x} - 2^{2x+3} + 12 = 0$, leaving your answer in terms of logarithms where necessary. (6 marks)

Problem solving: Set B

Practice Book, printed page 123.

Bronze

The value, V, in £s, of a mobile phone can be modelled by the formula $V = 50 + 700e^{-0.4t}$, $t \geq 0$, where t is the time in years since the phone was purchased.

a. Sketch the graph of V against t, labelling any asymptotes and any points of intersection with the coordinate axes. (3 marks)

b. Calculate the value of the phone after 5 years. (1 mark)

c. Find the time when the phone will be worth £100, giving your answer in the form $a \ln b$, where a and b are constants to be found. (3 marks)

Silver

The number of computers (in thousands), C, infected by an email virus can be modelled by the equation $C = e^{t/4} - 1$, $0 \leq t \leq 24$, where t is the time in hours since the email was sent.

a. Calculate the number of computers, to the nearest whole number, infected after 11 hours. (1 mark)

b. Find $\frac{dC}{dt}$ (2 marks)

c. Find the time when the number of computers being infected is increasing at a rate of 12000 per hour, leaving your answer in the form $a \ln b$, where a and b are constants to be found. (3 marks)

Gold

The temperature, $T\,°\text{C}$, of a bowl of soup is given by $T = ae^{kt} + 25$, $t \geq 0$, where t is the time in minutes since the soup was served.

a. Explain briefly why k must be negative. (1 mark)

Given that initially the temperature of the soup is 70°C and after 5 minutes the temperature is 55°C:

b. state the value of a (1 mark)

c. find the value of k in the form $c \ln d$, where c and d are constants to be found (3 marks)

d. find the time, to 2 decimal places, when the temperature of the soup is decreasing at a rate of 1.5°C per minute. (3 marks)

Selected exam question bank questions

Question 3

$f(x) = -2^x + 3$

Sketch the graph of $y = f(x)$, labelling the exact coordinates of any points of intersection with the coordinate axes and the equations of any asymptotes. (4 marks)

Question 18

Find the exact value of x such that $\log_{16}(1 - 3x) = \frac{3}{4}$ (3 marks)

Question 25

Find the exact solutions to the equation $6e^{6x} - 38e^{3x} + 40 = 0$ (5 marks)

Question 54

a. Given that $\log_3(7x - 18) - 2\log_3(x - 2) = 1$, show that $3x^2 - 19x + 30 = 0$. (4 marks)

b. Hence, or otherwise, solve $\log_3(7x - 18) - 2\log_3(x - 2) = 1$. (2 marks)

Question 59

The amount of money, A, in a bank account t years after it is deposited can be given by the equation $A = Pe^{rt/100}$, where P is the amount deposited and $r\%$ is the interest rate.

Given that £700 was deposited 4 years ago and that the interest rate on the account is 3.5%, find:

a. the current value of the account (2 marks)

b. $\frac{dA}{dt}$ (2 marks)

c. the time taken for the money to double. (3 marks)

Question 68

The number of reported cases of a disease in a country are shown in the table.

Year	**Cases,** C
1990	5128
1995	3943
2000	3031
2005	2331
2010	1792
2015	1378

This data can be modelled using an exponential function of the form $C = ab^t$, where t is the time in years since 1990 and a and b are constants.

a. Show that $C = ab^t$ can be written as $\log_{10} C = \log_{10} a + t \log_{10} b$. (2 marks)

b. Plot a graph of $\log_{10} C$ against t and draw in a line of best fit. (3 marks)

c. Use your graph to estimate the values of a and b to 2 significant figures. (3 marks)

Answers

14.1 Exponential functions

1.

a. $\frac{8}{27}$

b. $\frac{1}{16}$

c. 25

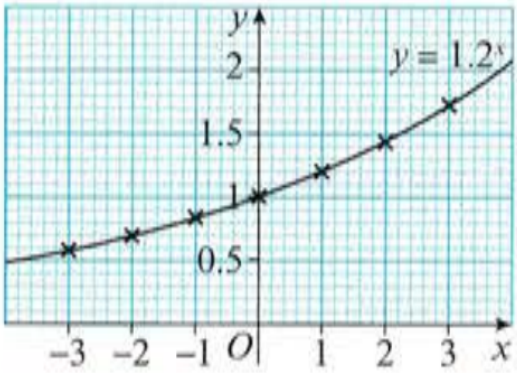
d. $\frac{27}{64}$

2.

a.

x	-3	-2	-1	0	1	2	3
y	0.58	0.69	0.83	1	1.2	1.44	1.73

b.

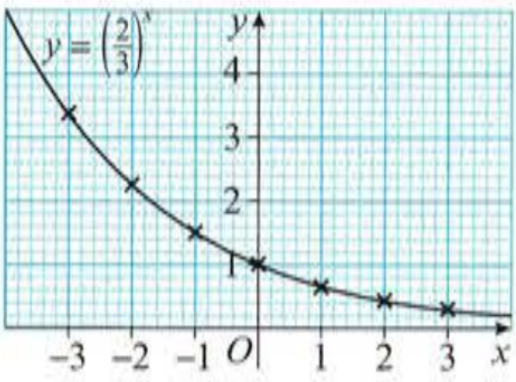


3.

a.

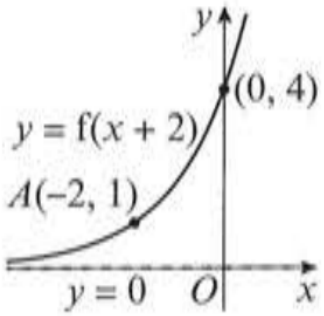
x	-3	-2	-1	0	1	2	3
y	$\frac{27}{8}$	$\frac{9}{4}$	$\frac{3}{2}$	1	$\frac{2}{3}$	$\frac{4}{9}$	$\frac{8}{27}$

b.

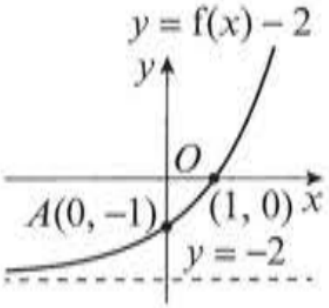


c. $x \approx -1.7$

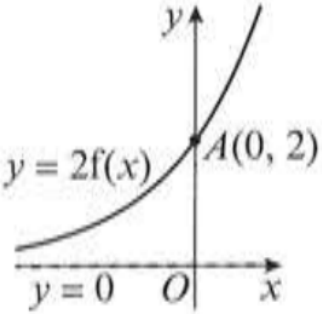
4. a



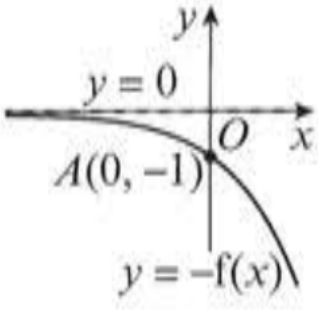
b



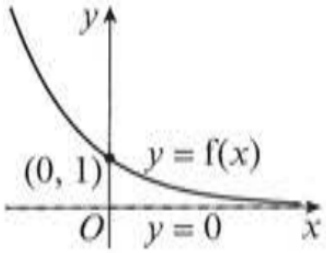
c



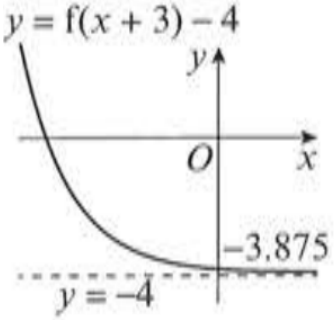
d



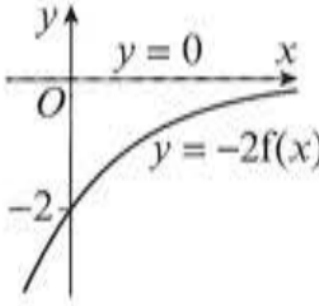
5. a



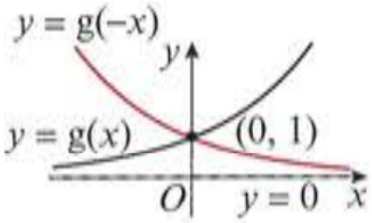
b i



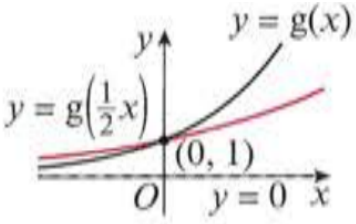
b ii



6. a.



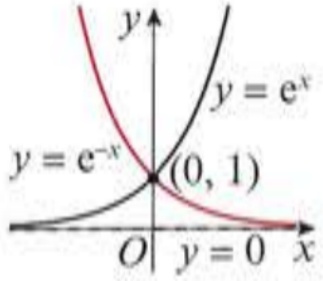
b.



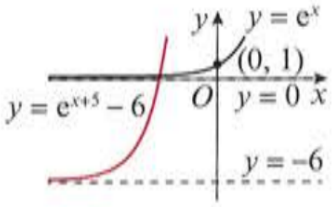
7. $k = 8$, $a = \frac{1}{4}$

14.2 $y = e^x$

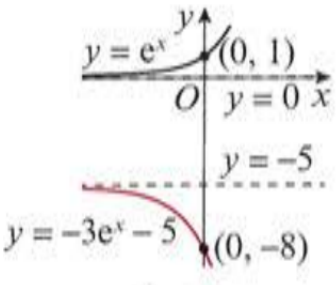
1. **a**



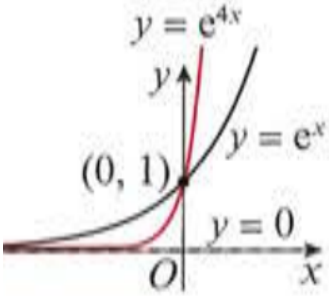
b



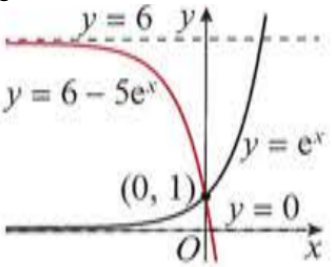
c



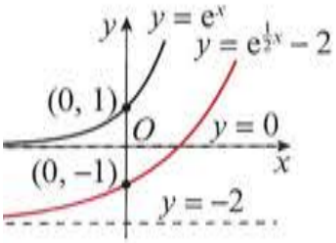
d



e



f



2.

a. $\frac{1}{2}e^{x/2}$

b. $-5e^{-5x}$

c. $-24e^{6x}$

d. $\frac{7}{3}e^{x/3}$

3.

a. $6e^{2x} + 4e^{4x}$

b. $-12e^{-2x} - 5e^{-x}$

c. $e^x - 8e^{2x}$

d. $-6e^{-3x} - 2e^{2x}$

4.

a. $A = e^4$, $b = 5$

b. $A = e^{-1}$, $b = 7$

c. $A = e^3$, $b = -\frac{1}{2}$

5.

a. $16x + \frac{6}{e^{6x}}$

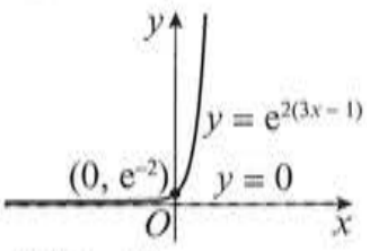
b. $32e^{4x} + 6e^{-x}$

6. $\frac{5}{2\sqrt{x}} - \frac{20}{x^3} + 8e^{-2x}$

7.

a. $e^{2(3x-1)} = e^{6x-2} = e^{6x} \times e^{-2} = e^{-2}(e^{6x})$, so $A = e^{-2}$, $b = 6$

b.



8. $y = 4e^2x - e^2$

14.3 Exponential modelling

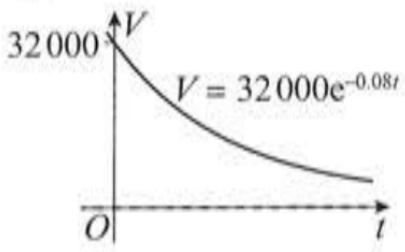
1.

a. $\frac{dP}{dt} = 200e^{0.01t}$

b. 221.0 (1 d.p.)

2.

a.



b. $\frac{dV}{dt} = -2560e^{-0.08t}$

c. -516.9 (1 d.p.)

d. The slope is negative, so the value of the car is decreasing.

3.

a.

i. 1300

ii. 1669

iii. 1777

b. It is the initial deer population.

c. $\frac{dD}{dt} = 162.5e^{t/8}$

d. Substitute $t = 8$ into $\frac{dD}{dt} = 162.5e^{t/8}$
$162.5e^1 = 441.72\ldots = 440$ (2 s.f.)

e. $\frac{dD}{dt} \approx 3579300$ which is not a reasonable value.

4.

a. £14115.82

b.

i. -2737.46

ii. -1693.90

c. The value decreases over the first 6 years of ownership. The decrease happens more quickly at first and then more slowly.

5.

a. $P_0 = 18000$

b. 594

c. For large values of t, the population will be increasing at an increasingly rapid rate, making it very unlikely the model is valid.

6.

a. 101.6 mg (1 d.p.)

b. $\frac{dP}{dt} = -0.15P$, so $k = -0.15$

c. k is negative, so the amount of paracetamol in the body is decreasing.

7.

a. £150000

b. £228294

c. $9000e^{0.06t}$

d. $t = 8$, $\frac{dV}{dt} = 14544.67$ and $t = 9$, $\frac{dV}{dt} = 15444.06$. During this time the property prices are increasing by £15000 per year, so the conditions are right for a crash.

14.4 Logarithms

1.
- **a.** $\log_{10} 1000 = 3$
- **b.** $2^3 = 8$
- **c.** $\log_7 49 = 2$
- **d.** $\log_6 \frac{1}{216} = -3$
- **e.** $4^{-4} = \frac{1}{256}$
- **f.** $9^1 = 9$
- **g.** $\log_{25} 5 = \frac{1}{2}$
- **h.** $27^{1/3} = 3$

2.
- **a.** 2
- **b.** -4
- **c.** 1
- **d.** $\frac{1}{3}$
- **e.** -2
- **f.** 0

3.
- **a.** 81
- **b.** 42
- **c.** 9
- **d.** 4

4.
- **a.** 1.7959
- **b.** -0.5108
- **c.** 2.1133

5. $\frac{33}{2}$

6. 256

7. $\frac{1}{625}$

8. $\frac{2}{3}$

14.5 Laws of logarithms

1.
- **a.** $\log_7 16$
- **b.** $\log_3 8$
- **c.** $\log_5 42$
- **d.** $\log_4 8000$
- **e.** $\log_7(\frac{64}{81})$
- **f.** $\log_6(\frac{12}{5})$

2.
- **a.** 2
- **b.** 2
- **c.** 3
- **d.** -1
- **e.** $\frac{2}{3}$

3.

a. 3

b. $2\log_a x + 3\log_a y$

c. $3\log_a x - 4\log_a z$

d. $\log_a x + \frac{1}{2}\log_a y - 2\log_a z$

e. $\frac{1}{3} + \frac{2}{3}\log_a x + \frac{1}{3}\log_a y + \frac{4}{3}\log_a z$

f. $3 + 4\log_a x + \log_a y - \frac{1}{2}\log_a z$

4.

a. 9

b. 288

c. 12

d. 7

5. $\frac{5}{8}$

6. $5\sqrt[3]{5}$

7. $\frac{7}{17}$

8.

a. $\log_2 \frac{x+5}{(x-1)^2} = 1$
$x + 5 = 2x^2 - 4x + 2$
$2x^2 - 5x - 3 = 0$

b. $x = 3$

14.6 Solving equations using logarithms

1.

a. 1.92

b. 3.87

c. 2.01

2.

a. -3.75

b. 4.10

c. -0.0581

3.

a. 1, 1.11

b. 1, 1.95

c. 0.431, 0.365

4.

a. $(3u + 4)(3u - 5) = 0$
$u = -\frac{4}{3}$, $u = \frac{5}{3}$

b. $3^{2x+2} - 3(3^x) - 20 = 0$
$3^2(3^{2x}) - 3(3^x) - 20 = 0$
$9(3^x)^2 - 3(3^x) - 20 = 0$
$9u^2 - 3u - 20 = 0$

c. $u = \frac{5}{3} \Rightarrow 3^x = \frac{5}{3} \Rightarrow x = \log_3 \frac{5}{3} = 0.465$
$u = -\frac{4}{3} \Rightarrow 3^x = -\frac{4}{3}$
This has no real solution as you cannot take the logarithm of a negative number

5.

a. 10.3

b. 1.41

c. 1.89

6.

a. 4.04

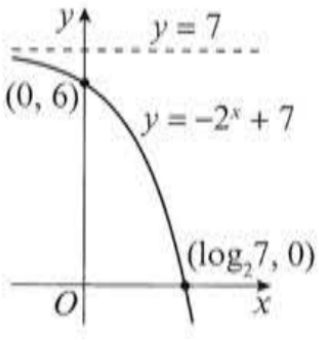
b. 0.613

7.

a. 19.6

b. 0.257

8. **a.**



b. 3.70

9.

a. $9^x - 3^{x+1} - 10 = 0$
$(3^2)^x - 3(3^x) - 10 = 0$
$(3^x)^2 - 3(3^x) - 10 = 0$
$u^2 - 3u - 10 = 0$

b. 1.46

14.7 Working with natural logarithms

1.

a. $\ln 8$
b. $2\ln 6$ or $\ln 36$
c. $-5 + \ln 81$
d. $\frac{1}{5} + \frac{1}{5}\ln 23$
e. $1 - \ln \frac{29}{4}$
f. $\frac{1}{3}\ln \frac{21}{2}$

2.

a. e^{11}
b. $\frac{1}{6}e^3$
c. $\frac{3e-1}{5e}$
d. $\frac{5-e^{2/5}}{2}$
e. $\frac{5+e^{7/2}}{3}$
f. 4, 5

3. $0, \ln 4$

4. $\frac{1}{2}\ln \frac{7}{2}, \frac{1}{2}\ln 2$

5. $\ln 16$

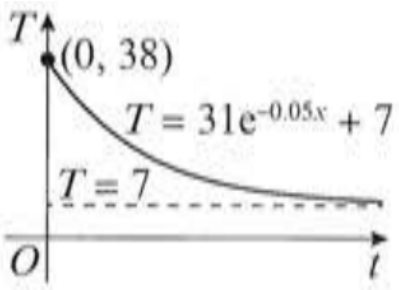
6. $\frac{1}{4}\ln \frac{3}{2}, \frac{1}{4}\ln 5$

7. $\frac{1-\ln 4}{6+\ln 12}$

8.

a. $(0, 3 - \ln 4)$
b. $(-2 + e^{3/2}, 0)$

9. **a.**



b. The minimum temperature of the bath water.

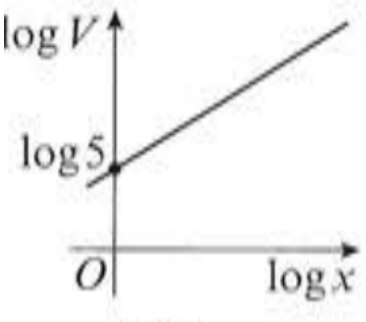
c. 10.9 minutes

14.8 Logarithms and non-linear data

1.

a. $V = 5x^3$
$\log V = \log 5x^3$
$\log V = \log 5 + 3\log x$

b.

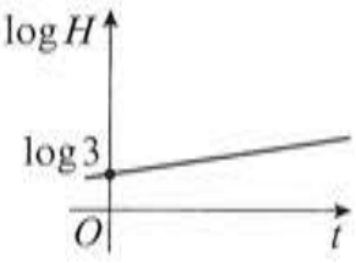


c. 3

2.

a. $H = 3(\frac{3}{2})^t$
$\log H = \log(3(\frac{3}{2})^t)$
$\log H = \log 3 + \log(\frac{3}{2})^t$
$\log H = \log 3 + t\log \frac{3}{2}$

b.



c. $\log \frac{3}{2}$

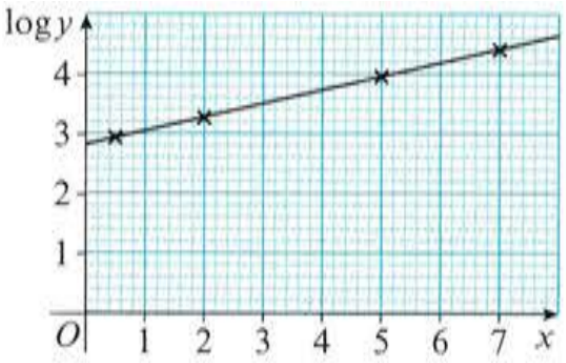
3.

a. $y = ab^x$
$\log y = \log(ab^x)$
$\log y = \log a + \log b^x$
$\log y = \log a + x \log b$

b.

x	0.5	2	5	7
$\log y$	2.93	3.27	3.97	4.43

c.



d. $a = 650$, $b = 1.7$

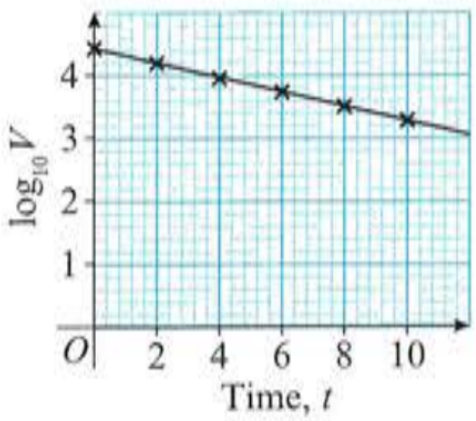
4.

a.

Time, t	0	2	4	6	8	10
$\log_{10} V$	4.40	4.17	3.94	3.72	3.49	3.26

b. $V = ab^t$
$\log V = \log(ab^t)$
$\log V = \log a + \log b^t$
$\log V = \log a + t \log b$

c.



d. $\frac{3.26-4.4}{10-0} = -0.114$

e. $a = 25100$, $b = 0.770$

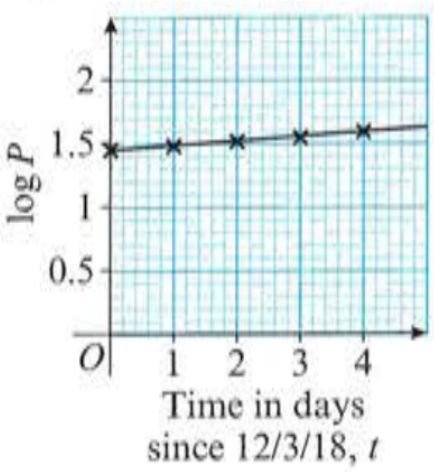
5.

a.

Time in days since 12/3/18, t	0	1	2	3	4
$\log P$	1.44	1.47	1.51	1.54	1.58

b. $P = ab^t$
$\log P = \log(ab^t)$
$\log P = \log a + \log b^t$
$\log P = \log a + t \log b$

c.



d. $a = 27.4$, $b = 1.09$

6.

a. $\log_4 P = -0.075t + 5.2$

b. 1350

c. $a = 1350$, $b = 0.901$

d. 60 people

7.

a. $\log_2 N = 0.26t + 9.3$

b. $a = 630.3$, $b = 1.2$

c. The initial number of bacteria in the petri dish.

Problem solving: Set A

Bronze

1.25

Silver

a. $9^x - 3^{x+2} + 20 = 0$
$(3^2)^x - (3^x)(3^2) + 20 = 0$
$(3^x)^2 - 9(3^x) + 20 = 0$
$u^2 - 9u + 20 = 0$

b. $\log_3 4$, $\log_3 5$

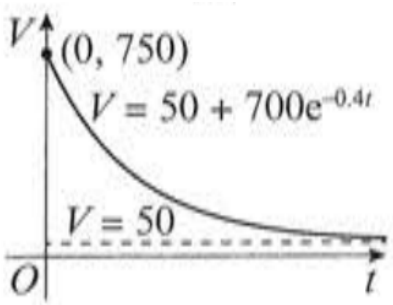
Gold

$\frac{1}{2}$, $\frac{\log 6}{2 \log 2}$

Problem solving: Set B

Bronze

a.



b. £144.73

c. $-\frac{5}{2}\ln\frac{1}{14}$ or $\frac{5}{2}\ln 14$

Silver

a. 14643 computers

b. $\frac{dC}{dt} = \frac{1}{4}e^{t/4}$

c. $4\ln 48$

Gold

a. Temperature of soup will decrease, so k must be negative.

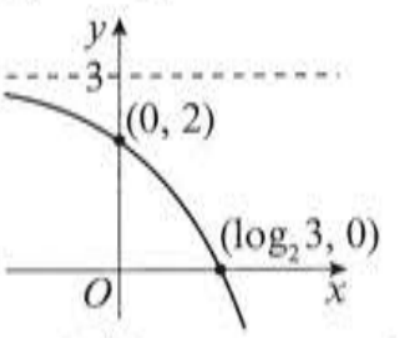
b. $a = 45$

c. $\frac{1}{5}\ln\frac{2}{3}$

d. 10.96 minutes

Selected exam question bank answers

Question 3



Question 18 $-\frac{7}{3}$

Question 25 $\frac{1}{3}\ln\frac{4}{3}$, $\frac{1}{3}\ln 5$

Question 54

a. $\log_3\frac{7x-18}{(x-2)^2} = 1$

$7x - 18 = 3x^2 - 12x + 12$

$3x^2 - 19x + 30 = 0$

b. $x = 3$, $x = \frac{10}{3}$

Question 59

a. £805.19

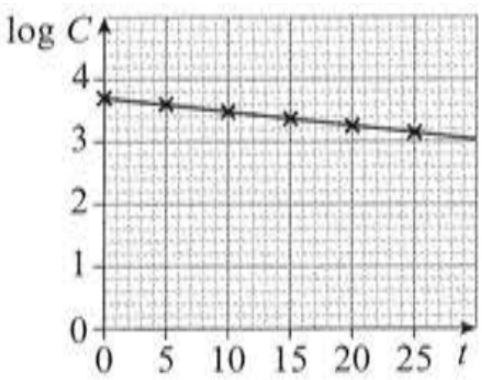
b. $24.5e^{0.035t}$

c. 19.8 years

Question 68

a. $C = ab^t$
$\log C = \log(ab^t)$
$\log C = \log a + \log b^t$
$\log C = \log a + t \log b$

b.



c. $a = 5100$, $b = 0.95$